

Iris Flower Classification using K-Nearest Neighbours

(KNN)

Aim

To implement the **K-Nearest Neighbours (KNN)** algorithm in Python for classifying Iris flowers into different species.

Apparatus Required

- Google Colab / Jupyter Notebook
- Python 3
- Libraries:
 - NumPy
 - Pandas
 - Scikit-learn (sklearn)

Theory

The **K-Nearest Neighbours (KNN)** algorithm is a supervised machine learning algorithm used for classification and regression. It classifies a new data sample based on the majority class among its **K nearest neighbours** using a distance metric such as Euclidean distance.

The **Iris dataset** contains 150 flower samples belonging to three species:

- Iris Setosa
- Iris Versicolor
- Iris Virginica

Each sample has four features:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width

Algorithm

1. Import the required libraries.
2. Load the Iris dataset.
3. Split the dataset into training and testing sets.
4. Standardize the feature values.
5. Train the KNN classifier.
6. Predict the species of the test samples.
7. Display the Accuracy and Confusion Matrix.

Code (Google Colab)

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Load Iris dataset
iris = load_iris()
X = iris.data
y = iris.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
```

```
# Feature Scaling
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# KNN Model
knn = KNeighborsClassifier(n_neighbors=3)
knn.fit(X_train, y_train)

# Prediction
y_pred = knn.predict(X_test)

# Results
print("Predicted Classes:")
print(y_pred)

print("\nAccuracy:")
print(accuracy_score(y_test, y_pred))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=iris.target_names))
```

Sample Output

```

*** Predicted Classes:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 1
 0 0 0 2 1 1 0 0]

Accuracy:
1.0

Confusion Matrix:
[[19  0  0]
 [ 0 13  0]
 [ 0  0 13]]

Classification Report:

```

	precision	recall	f1-score	support
setosa	1.00	1.00	1.00	19
versicolor	1.00	1.00	1.00	13
virginica	1.00	1.00	1.00	13
accuracy			1.00	45
macro avg	1.00	1.00	1.00	45
weighted avg	1.00	1.00	1.00	45

Result

The **Iris Flower Classification** was successfully implemented using the **K-Nearest Neighbours (KNN)** algorithm in Python. The model accurately classified the Iris flowers into **Setosa, Versicolor, and Virginica**, and its performance was evaluated using **Accuracy, Confusion Matrix, and the Classification Report**.